**Design e Arquitectura de Software**

**2022/2023**

**Lab. Classes - 4**

**GoF - Behavioral Patterns**

1. Consider a class Snapshot that stores internally a string. This class allows its internal state to be represented using the **Memento** pattern. Implement a program that uses a list of Mementos to manipulate the state of a snapshot object.

```
The current state is Snapshot #0. Saving this checkpoint.
The current state is Snapshot #1. Saving this checkpoint.
The current state is Snapshot #2. Saving this checkpoint.
The current state is Snapshot #3. Saving this checkpoint.
The current state is Snapshot #4
Restored to state: Snapshot #3
Restored to state: Snapshot #2
Restored to state: Snapshot #1
Restored to state: Snapshot #0
Res   tored to state: Snapshot #3
```

2. Implement the **Strategy** pattern, as illustrated in slide nr. 15.
   An example of expected output is:

```
The airplane is in born state.
It cannot do anything special.
Setting flying capability to it.
The airplane can fly now.
Changing the vehicle behavior again.
The airplane can float now.
```

3. Implement the **state** pattern example illustrated in slide nr. 21.
   An example of expected output is:

```
Creating new connection. Initial state is Closed.
Passive opening. State is now Listen.
Acknowledging. State is now Established.
Closing. State is now Listen.
Active opening. Unexpected message. State is now Listen.
Closing. Unexpected message. State is now Listen.
Acknowledging. State is now Established.
```

4. Implement the **mediator** pattern example illustrated in slides nr. 26 and 27 (you can ignore the Button object).

   An example of use would be:

```
Client::Creating Font Dialog Director.
Director::Creating List Box with options Op1, Op2, Op3
Director::Creating Entry Field with empty text.
Client:: Acessing listbox to select new option
```

```
Listbox:: Listbox changed to Op1. Notifying director.
Director:: Director knows Listbox changed. Retrieving selected
option Op1.
Director::Director setting Entry Field text to Op1.
Entry Field::Entry Field text set to Op1.
Client:: Acessing listbox to select new option
ListBox::Listbox changed to Op2. Notifying director.
Director::Director knows Listbox changed. Retrieving selected
option Op2.
Director::Director setting Entry Field text to Op2.
EntryField::Entry Field text set to Op2.
```

5. Implement the **Visitor** pattern, considering the example of slide nr. 33.
   An example of use would be:

```
Testing the IncrementorVisitor now.
The flag value is:5
Incrementing it by 10.
The new value is:15
Testing the DoubleMakerVisitor now.
The flag value is:5
Multiplying it by 2.
The new value is:10
```

6. Implement the **Interpreter** pattern, considering the example of slide nr. 37.

Suggestion: You can use the `StringTokenizer` class or `split()` method to break up the string into elementary tokens.